

Kubernetes-09

Tasks:

1. Execute all the steps shown in video.
2. Upgrade k8s cluster from 1.26 to 1.28 Version. Note: Document to upgrade cluster. <https://devopscube.com/upgrade-kubernetes-cluster-kubeadm/>

Task: 1

1. Execute all the steps shown in video.

Ingress :

Ingress:

=====

Ingress will help us to expose the applications and manage external access by providing http/https routing rules to the services within a k8s cluster.

Ingress controller:

=====

Nothing but a load balancer which is responsible for implementing and managing Ingress resources.

We have multiple load balancers in the market like

NGINX

A10

F5

Traefik

HA PROXY

Ingress is developed by k8s.

Ingress controller is developed by the load balancer companies.

Why should we use Ingress?

prior to ingress we were using load balancer service to expose our application to outside world, but it only provides round robin algorithm.

There were couple of issues for applications, for example if we have 100 services then we need to create 100 services. It will not support different features which our load balancer's were doing. Like path based routing, rules, whitelisting, blacklisting, tls, sticky session, host based routing.

To overcome the problem Ingress has been introduced to k8s.

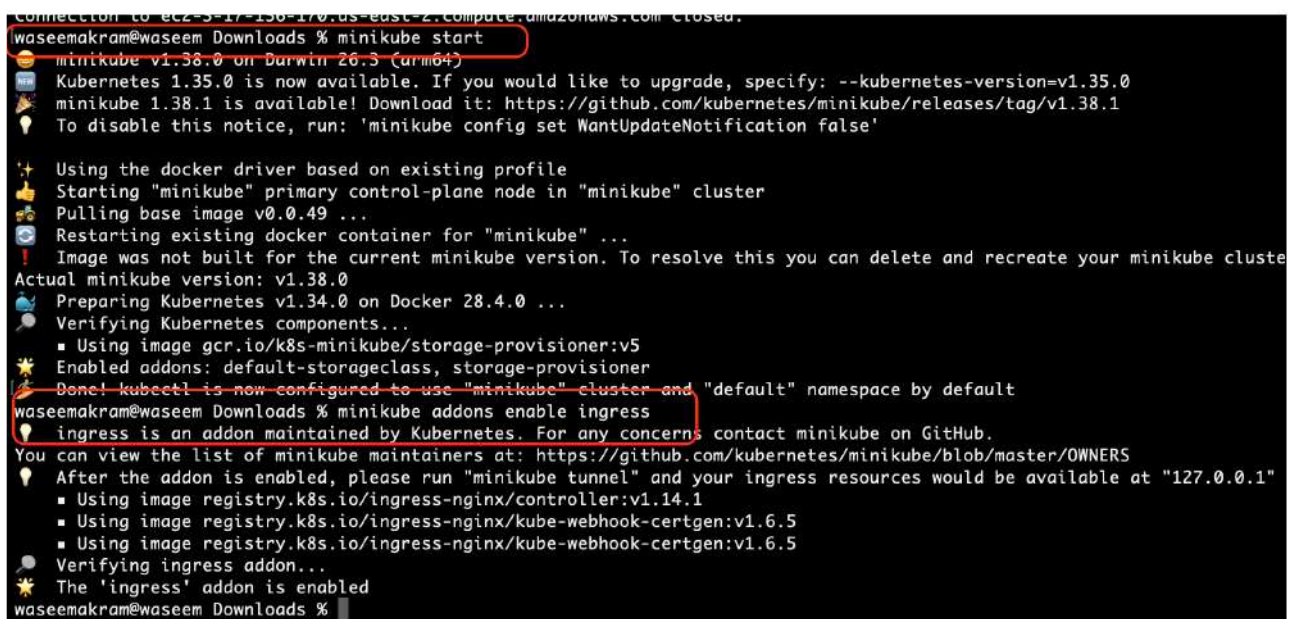
What needs to be done by user?

- 1) He needs to create one ingress.
- 2) He needs to create ingress controller based upon the features/capabilities we need.

STEP 1 – Start Minikube

```
minikube start
```

```
minikube addons enable ingress
```



```
waseemakram@waseem Downloads % minikube start
minikube v1.38.0 on Darwin 26.3 (arm64)
Kubernetes 1.35.0 is now available. If you would like to upgrade, specify: --kubernetes-version=v1.35.0
minikube 1.38.1 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.38.1
To disable this notice, run: 'minikube config set WantUpdateNotification false'

* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.49 ...
* Restarting existing docker container for "minikube" ...
! Image was not built for the current minikube version. To resolve this you can delete and recreate your minikube cluster
Actual minikube version: v1.38.0
* Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
waseemakram@waseem Downloads % minikube addons enable ingress
! ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
* After the addon is enabled, please run "minikube tunnel" and your ingress resources would be available at "127.0.0.1"
  - Using image registry.k8s.io/ingress-nginx/controller:v1.14.1
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.5
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.5
* Verifying ingress addon...
* The 'ingress' addon is enabled
waseemakram@waseem Downloads %
```

Check ingress pods:

```
kubectl get pods -A
```

You must see:

```
waseemakram@waseem Downloads % kubectl get pods -A
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
ingress-nginx	ingress-nginx-admission-create-wl9f9	0/1	Completed	0	2m21s
ingress-nginx	ingress-nginx-admission-patch-t4tr7	0/1	Completed	1	2m21s
ingress-nginx	ingress-nginx-controller-85d4c799dd-hdrhg	1/1	Running	0	2m21s
kube-system	coredns-66bc5c9577-kfn4k	1/1	Running	1 (9d ago)	10d
kube-system	etcd-minikube	1/1	Running	1 (9d ago)	10d
kube-system	kube-apiserver-minikube	1/1	Running	1 (9d ago)	10d
kube-system	kube-controller-manager-minikube	1/1	Running	1 (9d ago)	10d
kube-system	kube-proxy-thrqh	1/1	Running	1 (9d ago)	10d
kube-system	kube-scheduler-minikube	1/1	Running	1 (9d ago)	10d
kube-system	storage-provisioner	1/1	Running	3 (7m30s ago)	10d

```
waseemakram@waseem Downloads %
```

ingress-nginx-controller Running

STEP 2 – Create Deployment

```
vi deployment.yml
```

```
kube-system      storage-provisioner
waseemakram@waseem Downloads % vi deployment.yml
waseemakram@waseem Downloads % cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
waseemakram@waseem Downloads %
```

Save and apply:

```
kubectl apply -f deployment.yml  
kubectl get pods
```

Wait until:

1/1 Running

```
- containerPort: 80  
waseemakram@waseem Downloads % kubectl apply -f deployment.yml  
  
deployment.apps/nginx-deployment created  
waseemakram@waseem Downloads % kubectl get pods  
NAME                                READY   STATUS             RESTARTS   AGE  
nginx-deployment-6f9664446b-h8dfz   0/1     ContainerCreating   0           10s  
waseemakram@waseem Downloads % kubectl get pods  
NAME                                READY   STATUS    RESTARTS   AGE  
nginx-deployment-6f9664446b-h8dfz   1/1     Running   0           20s  
waseemakram@waseem Downloads %
```

STEP 3 – Create Service (ClusterIP)

Important: For Ingress → use ClusterIP

Create:

```
vi service.yml
```

```
waseemakram@waseem Downloads % vi service.yml

[waseemakram@waseem Downloads % cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - port: 80
    targetPort: 80
  type: ClusterIP
waseemakram@waseem Downloads % █
```

Apply:

```
kubectl apply -f service.yml
kubectl get svc
```

You should see:

```
nginx-service    ClusterIP
```

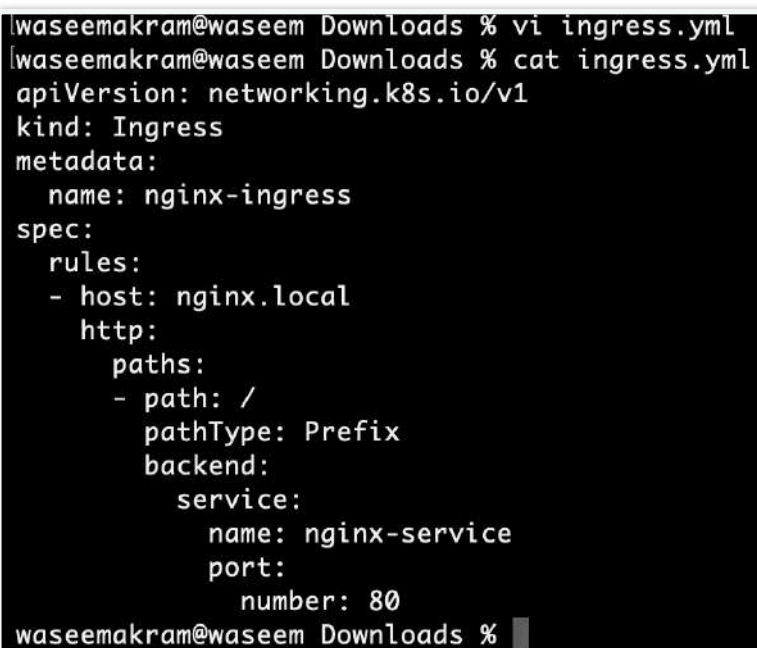
```
type: ClusterIP
waseemakram@waseem Downloads % kubectl apply -f service.yml

service/nginx-service created
[waseemakram@waseem Downloads % kubectl get psvc
error: the server doesn't have a resource type "psvc"
[waseemakram@waseem Downloads % kubectl get svc
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes          ClusterIP     10.96.0.1     <none>         443/TCP    10d
nginx-service       ClusterIP     10.98.68.48   <none>         80/TCP     11s
waseemakram@waseem Downloads % █
```

STEP 4 – Create Ingress (Use Your Domain)

Create:

```
vi ingress.yml
```

A terminal window with a black background and white text. The prompt is 'waseemakram@waseem Downloads %'. The user enters 'vi ingress.yml' and then 'cat ingress.yml'. The output shows the contents of the ingress.yml file, which defines an Ingress resource named 'nginx-ingress' that routes traffic from 'nginx.local' to 'nginx-service' on port 80.

```
waseemakram@waseem Downloads % vi ingress.yml
waseemakram@waseem Downloads % cat ingress.yml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-ingress
spec:
  rules:
  - host: nginx.local
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: nginx-service
            port:
              number: 80
waseemakram@waseem Downloads %
```

Apply:

```
kubectl apply -f ingress.yml
kubectl get ingress
```

```
number: 80
[waseemakram@waseem Downloads % kubectl apply -f ingress.yml
ingress.networking.k8s.io/nginx-ingress configured
[waseemakram@waseem Downloads % kubectl get ingress
NAME          CLASS   HOSTS          ADDRESS        PORTS   AGE
nginx-ingress  nginx   nginx.local    192.168.49.2   80      24m
waseemakram@waseem Downloads %
```

STEP 5 – Start Minikube Tunnel (VERY

IMPORTANT)

Run:

```
sudo minikube tunnel
```

Enter your Mac password

```
waseemakram@waseem Downloads % sudo minikube tunnel
password:
Tunnel successfully started

NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ..

The service/ingress nginx-ingress requires privileged ports to be exposed: [80 443]
sudo permission will be asked for it.
Starting tunnel for service nginx-ingress.
```

 **Keep this terminal OPEN**

Open new terminal and check:

```
kubectl get ingress
```

STEP 6 – Edit Hosts File (MacOS)

Now map your domain locally.

Open:

```
sudo vi /etc/hosts
```

Go to bottom and add:

```
##
# Host Database
#
# localhost is used to configure the loopback interface
# when the system is booting. Do not change this entry.
##
127.0.0.1        localhost
255.255.255.255 broadcasthost
::1             localhost
127.0.0.1        nginx.local
~
~
~
~
~
```

```
waseemakram@waseem Downloads % sudo vi /etc/hosts
```

```
[Password:
```

```
waseemakram@waseem Downloads %
```

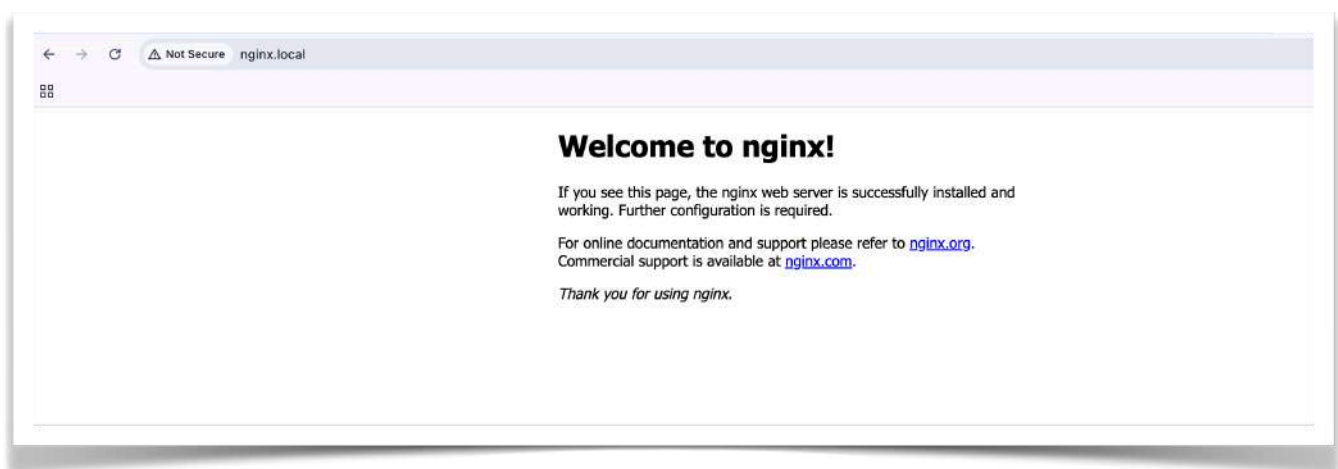

NOW OPEN BROWSER

Type:

`http://nginx.local`

You should see:

`Welcome to nginx!`



You just successfully implemented:

Browser → Domain → Ingress → Service → Pod

Task:2

2.Upgrade k8s cluster from 1.26 to 1.28 Version. Note: Document to upgrade cluster. <https://devopscube.com/upgrade-kubernetes-cluster-kubeadm/>

Step 1: Check Existing kubeadm Version

```
kubeadm version -o json
```

Confirm it shows 1.30.x

```
wait
[ubuntu@master:~$ kubectl get nodes
NAME          STATUS    ROLES          AGE   VERSION
master        Ready     control-plane   9d    v1.30.14
worker1       Ready     <none>          9d    v1.30.14
worker2       Ready     <none>          9d    v1.30.14
[ubuntu@master:~$ kubeadm version -o json
{
  "clientVersion": {
    "major": "1",
    "minor": "30",
    "gitVersion": "v1.30.14",
    "gitCommit": "9e18483918821121abdf9aa82bc14d66df5d68cd",
    "gitTreeState": "clean",
    "buildDate": "2025-06-17T18:34:53Z",
    "goVersion": "go1.23.10",
    "compiler": "gc",
    "platform": "linux/amd64"
  }
}
[ubuntu@master:~$
```

Step 2: Unhold kubeadm

During the Kubeadm cluster installation process, you would have hold the kubeadm installation to prevent upgrades.

Now we need to unhold kubeadm.

```
sudo apt-mark unhold kubeadm
```

Check the available latest Kubeadm version using the following command.

```
sudo apt-cache madison kubeadm | tac
```

```
ubuntu@master:~$ sudo apt-mark unhold kubeadm
Canceled hold on kubeadm.
ubuntu@master:~$ sudo apt-cache madison kubeadm | tac
kubeadm | 1.30.0-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.1-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.2-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.3-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.4-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.5-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.6-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.7-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.8-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.9-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.10-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.11-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.12-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.13-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
kubeadm | 1.30.14-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
ubuntu@master:~$
```

That means:

- 👉 You only have 1.30 packages available
- 👉 1.31 is NOT available in your current repo

That's why `apt-cache madison kubeadm` is showing only 1.30.x.

Important

To upgrade to 1.31, you must change Kubernetes apt repo from:

`v1.30`

to

`v1.31`

Otherwise upgrade is impossible.

FIX — Add 1.31 Repository

On MASTER, run:

Add new repo for 1.31

```
echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-  
keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /"  
| sudo tee /etc/apt/sources.list.d/kubernetes.list
```

```

kubeadm | 1.30.14-1.1 | https://pkgs.k8s.io/core:/stable:/v1.30/deb Packages
ubuntu@master:~$ echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /
ubuntu@master:~$ sudo apt-get update
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1800 kB]
Get:4 https://prod-cdn.packages.k8s.io/repositories/ipv:kubernetes:/core:/stable:/v1.31/deb InRelease [1192 B]
Get:6 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:7 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.5 kB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1557 kB]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [32.7 kB]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:14 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7300 B]
Get:15 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:16 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:17 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:18 https://prod-cdn.packages.k8s.io/repositories/ipv:kubernetes:/core:/stable:/v1.31/deb Packages [20.8 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1491 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9932 B]
Get:22 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [937 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [217 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.1 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:27 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Fetched 7158 kB in 2s (4457 kB/s)
Reading package lists... Done
ubuntu@master:~$

```

—> Update packages

```
sudo apt-get update
```

—> Now check available versions

```
sudo apt-cache madison kubeadm | tac
```

Now you should see:

```
1.31.x-1.1
```

```
ubuntu@master:~$ sudo apt-cache madison kubeadm | tac
kubeadm | 1.31.0-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.1-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.2-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.3-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.4-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.5-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.6-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.7-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.8-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.9-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.10-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.11-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.12-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.13-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
kubeadm | 1.31.14-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb Packages
ubuntu@master:~$
```

After That

Install latest 1.31 version (example):

Step 1 — Install kubeadm 1.31.14

```
sudo apt-get install -y kubeadm=1.31.14-1.1
```

Then hold it:

```
sudo apt-mark hold kubeadm
```



```

kubeadm v1.31.14-1.1 | https://pkg.k8s.io/core/stable/v1.31/deb Packages
ubuntu@master:~$ sudo apt-get install -y kubeadm=1.31.14-1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be upgraded:
  kubeadm
1 upgraded, 0 newly installed, 0 to remove and 56 not upgraded.
Need to get 11.6 MB of archives.
After this operation, 8618 kB of additional disk space will be used.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isv/kubernetes:/core:/stable:/v1.31/deb kubeadm 1.31.14-1.1 [11.6 MB]
Fetched 11.6 MB in 2s (7716 kB/s)
(Reading database ... 104132 files and directories currently installed.)
Preparing to unpack .../kubeadm_1.31.14-1.1_amd64.deb ...
Unpacking kubeadm (1.31.14-1.1) over (1.30.14-1.1) ...
Setting up kubeadm (1.31.14-1.1) ...
Scanning processes...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.14.0-1018-aws
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.17.0-1007-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@master:~$

```

`sudo apt-mark hold kubeadm`

Verify:

`kubeadm version`

It should show 1.31.x

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@master:~$ sudo apt-mark hold kubeadm
kubeadm set on hold.
ubuntu@master:~$ kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"31", GitVersion:"v1.31.14",
GoVersion:"go1.24.9", Compiler:"gc", Platform:"linux/amd64"}
ubuntu@master:~$

```

Step 2 — Check Upgrade Plan

```
sudo kubeadm upgrade plan
```

It should show:

Upgrade to v1.31.14

```
ubuntu@master:~$ sudo kubeadm upgrade plan
[preflight] Running pre-flight checks.
[upgrade/config] Reading configuration from the cluster...
[upgrade/config] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[upgrade] Running cluster health checks
[upgrade] Fetching available versions to upgrade to
[upgrade/versions] Cluster version: v1.30.14
[upgrade/versions] kubeadm version: v1.31.14
I0226 11:10:42.809739 27526 version.go:261] remote version is much newer: v1.35.1; falling back to: stable-1.31
[upgrade/versions] Target version: v1.31.14
[upgrade/versions] Latest version in the v1.30 series: v1.30.14

Components that must be upgraded manually after you have upgraded the control plane with 'kubeadm upgrade apply':
COMPONENT  NODE    CURRENT  TARGET
kubelet    master  v1.30.14 v1.31.14
kubelet    worker1 v1.30.14 v1.31.14
kubelet    worker2 v1.30.14 v1.31.14

Upgrade to the latest stable version:

COMPONENT            NODE    CURRENT  TARGET
kube-apiserver        master  v1.30.14 v1.31.14
kube-controller-manager master  v1.30.14 v1.31.14
kube-scheduler        master  v1.30.14 v1.31.14
kube-proxy            1.30.14 v1.31.14
CoreDNS               v1.11.3 v1.11.3
etcd                  master  3.5.15-0 3.5.24-0

You can now apply the upgrade by executing the following command:

    kubeadm upgrade apply v1.31.14

-----

The table below shows the current state of component configs as understood by this version of kubeadm.
Configs that have a "yes" mark in the "MANUAL UPGRADE REQUIRED" column require manual config upgrade or
resetting to kubeadm defaults before a successful upgrade can be performed. The version to manually
upgrade to is denoted in the "PREFERRED VERSION" column.

API GROUP            CURRENT VERSION  PREFERRED VERSION  MANUAL UPGRADE REQUIRED
kubeproxy.config.k8s.io v1alpha1        v1alpha1           no
kubelet.config.k8s.io  v1beta1         v1beta1            no
```


Step 3 — Apply Upgrade

```
sudo kubeadm upgrade apply v1.31.14
```

Wait until it finishes successfully.

```
ubuntu@master:~$ sudo kubeadm upgrade apply v1.31.14
[preflight] Running pre-flight checks.
[upgrade/config] Reading configuration from the cluster...
[upgrade/config] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[upgrade] Running cluster health checks
[upgrade/version] You have chosen to change the cluster version to "v1.31.14"
[upgrade/versions] Cluster version: v1.30.14
[upgrade/versions] kubeadm version: v1.31.14
[upgrade] Are you sure you want to proceed? [y/N]: y
[upgrade/prepull] Pulling images required for setting up a Kubernetes cluster
[upgrade/prepull] This might take a minute or two, depending on the speed of your internet connection
[upgrade/prepull] You can also perform this action beforehand using 'kubeadm config images pull'
W0226 11:12:50.250935 28751 checks.go:843] detected that the sandbox image "registry.k8s.io/pause:3.8" of the control plane is outdated. It is recommended to use "registry.k8s.io/pause:3.10" as the CRI sandbox image.
[upgrade/apply] Upgrading your Static Pod-hosted control plane to version "v1.31.14" (timeout: 5m0s)...
[upgrade/staticpods] Writing new Static Pod manifests to "/etc/kubernetes/tmp/kubeadm-upgraded-manifests3615241830"
[upgrade/staticpods] Preparing for "etcd" upgrade
[upgrade/staticpods] Renewing etcd-server certificate
[upgrade/staticpods] Renewing etcd-peer certificate
[upgrade/staticpods] Renewing etcd-healthcheck-client certificate
[upgrade/staticpods] Moving new manifest to "/etc/kubernetes/manifests/etcd.yaml" and backing up old manifest to "/etc/kubernetes/manifests/etcd.yaml.bak"
[upgrade/staticpods] Waiting for the kubelet to restart the component
[upgrade/staticpods] This can take up to 5m0s
```

```
[addons] Applied essential addon: CoreDNS
[addons] Applied essential addon: kube-proxy

[upgrade/successful] SUCCESS! Your cluster was upgraded to "v1.31.14". Enjoy!

[upgrade/kubelet] Now that your control plane is upgraded, please proceed with upgrading your kubelets...
ubuntu@master:~$
```

Step 4 — Drain Control Plane

Your node name is:

master

So run:

```
kubectldrain master --ignore-daemonsets --delete-emptydir-data
```

Step 5 — Upgrade kubelet & kubectldrain

```
sudo apt-mark unhold kubelet kubectldrain
sudo apt-get install -y kubelet=1.31.14-1.1
kubectldrain=1.31.14-1.1
sudo apt-mark hold kubelet kubectldrain
```

```
[Upgrade:kubectldrain] Now that your control plane is upgraded, please proceed with upgrading your kubelets if you haven't already done so.
ubuntu@master:~$ kubectldrain master --ignore-daemonsets --delete-emptydir-data
node/master cordoned
Warning: ignoring DaemonSet-managed Pods: kube-system/calico-node-nrkps, kube-system/ebs-csi-node-6vdjq, kube-system/efs-csi-node-rrfgx, k
evicting pod kube-system/coredns-55cb58b774-57jgs
evicting pod kube-system/calico-kube-controllers-5b9b456c66-m9q5l
evicting pod kube-system/coredns-55cb58b774-2xs8s
pod/calico-kube-controllers-5b9b456c66-m9q5l evicted
pod/coredns-55cb58b774-2xs8s evicted
pod/coredns-55cb58b774-57jgs evicted
node/master drained
ubuntu@master:~$ sudo apt-mark unhold kubelet kubectldrain
sudo apt-get install -y kubelet=1.31.14-1.1 kubectldrain=1.31.14-1.1
sudo apt-mark hold kubelet kubectldrain
Canceled hold on kubelet.
Canceled hold on kubectldrain.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  conntrack
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  kubectldrain kubelet
2 upgraded, 0 newly installed, 0 to remove and 54 not upgraded.
Need to get 26.9 MB of archives.
After this operation, 16.9 MB disk space will be freed.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.31/deb kubectldrain 1.31.14-1.1 [11.5 MB]
Get:2 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.31/deb kubelet 1.31.14-1.1 [15.4 MB]
```

Restart kubelet:

```
sudo systemctl daemon-reload  
sudo systemctl restart kubelet
```

Step 6 — Uncordon Master

```
kubectl uncordon master
```

```
kubectl set on hold.  
ubuntu@master:~$ sudo systemctl daemon-reload  
[sudo systemctl restart kubelet  
ubuntu@master:~$ kubectl uncordon master  
node/master uncordoned
```

Step 7 — Verify

```
kubectl get nodes
```

Expected result:

```
ubuntu@master:~$ kubectl get nodes  
NAME          STATUS    ROLES          AGE    VERSION  
master        Ready     control-plane   9d     v1.31.14  
worker1       Ready     <none>          9d     v1.30.14  
worker2       Ready     <none>          9d     v1.30.14  
ubuntu@master:~$
```

```
master    v1.31.14
worker1   v1.30.14
worker2   v1.30.14
```

This is correct. Workers still old version.

UPGRADE WORKER1

STEP 1 — Add 1.31 repo

Run:

```
echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-
keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ ."
| sudo tee /etc/apt/sources.list.d/kubernetes.list
```

STEP 2 — Update package list

```
sudo apt-get update
```

STEP 3 — Check kubeadm versions

```
sudo apt-cache madison kubeadm | tac
```

You should now see:

1.31.14-1.1

```
ubuntu@worker1:~$ echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/st
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /
ubuntu@worker1:~$ sudo apt-get update
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.31/deb InRelease [1192 B]
Get:6 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.31/deb Packages [20.8 kB]
Fetched 22.0 kB in 0s (90.1 kB/s)
Reading package lists... Done
ubuntu@worker1:~$ sudo apt-cache madison kubeadm | tac
  kubeadm | 1.31.0-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.1-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.2-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.3-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.4-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.5-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.6-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.7-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.8-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.9-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.10-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.11-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.12-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.13-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
  kubeadm | 1.31.14-1.1 | https://pkgs.k8s.io/core:/stable:/v1.31/deb | Packages
ubuntu@worker1:~$
```

i-0b424c636f70f9196 (k8s-worker1)

STEP 4 — Install kubeadm 1.31.14

```
sudo apt-mark unhold kubeadm
sudo apt-get install -y kubeadm=1.31.14-1.1
sudo apt-mark hold kubeadm
```

```
ubuntu@worker1:~$ sudo apt-mark unhold kubeadm
sudo apt-get install -y kubeadm=1.31.14-1.1
sudo apt-mark hold kubeadm
Canceled hold on kubeadm.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be upgraded:
  kubeadm
1 upgraded, 0 newly installed, 0 to remove and 65 not upgraded.
Need to get 11.6 MB of archives.
After this operation, 8618 kB of additional disk space will be used.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.31/deb kubeadm 1.31.14-1.1 [11.6 MB]
Fetched 11.6 MB in 0s (50.3 MB/s)
(Reading database ... 71875 files and directories currently installed.)
Preparing to unpack .../kubeadm_1.31.14-1.1_amd64.deb ...
Unpacking kubeadm (1.31.14-1.1) over (1.30.14-1.1) ...
Setting up kubeadm (1.31.14-1.1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kubeadm set on hold.
ubuntu@worker1:~$
```

i-0b424c636f70f9196 (k8s-worker1)

Check:

kubeadm version

It must show 1.31.14.

```
ubuntu@worker1:~$ kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"31", GitVersion:"v1.31.14", GitCommit:"f38478c547e971925a3a7e837f284f688196064", Compiler:"gc", Platform:"linux/amd64"}
ubuntu@worker1:~$
```

After this we continue with:

sudo kubeadm upgrade node

STEP 5 — Run kubeadm upgrade node

sudo kubeadm upgrade node

```
ubuntu@worker1:~$ sudo kubeadm upgrade node
[upgrade] Reading configuration from the cluster...
[upgrade] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[preflight] Running pre-flight checks
[preflight] Skipping prepull. Not a control plane node.
[upgrade] Skipping phase. Not a control plane node.
[upgrade] Skipping phase. Not a control plane node.
[upgrade] Backing up kubelet config file to /etc/kubernetes/tmp/kubeadm-kubelet-config437666935/config.yaml
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[upgrade] The configuration for this node was successfully updated!
[upgrade] Now you should go ahead and upgrade the kubelet package using your package manager.
ubuntu@worker1:~$
```


STEP 6 — Upgrade kubelet & kubectl

Run:

```
sudo apt-mark unhold kubelet kubectl
sudo apt-get install -y kubelet=1.31.14-1.1
kubectl=1.31.14-1.1
sudo apt-mark hold kubelet kubectl
```

```
ubuntu@worker1:~$ sudo apt-mark unhold kubelet kubectl
sudo apt-get install -y kubelet=1.31.14-1.1 kubectl=1.31.14-1.1
sudo apt-mark hold kubelet kubectl
Canceled hold on kubelet.
Canceled hold on kubectl.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  conntrack
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  kubectl kubelet
2 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
Need to get 26.9 MB of archives.
After this operation, 16.9 MB disk space will be freed.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.31/deb
Get:2 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.31/deb
Fetched 26.9 MB in 0s (76.0 MB/s)
(Reading database ... 71875 files and directories currently installed.)
Preparing to unpack .../kubectl_1.31.14-1.1_amd64.deb ...
Unpacking kubectl (1.31.14-1.1) over (1.30.14-1.1) ...
Preparing to unpack .../kubelet_1.31.14-1.1_amd64.deb ...
```


STEP 7 — Restart kubelet

```
sudo systemctl daemon-reload  
sudo systemctl restart kubelet
```

```
ubuntu@worker1:~$ sudo systemctl daemon-reload  
sudo systemctl restart kubelet  
ubuntu@worker1:~$
```

STEP 8 — Verify

```
kubectl get nodes
```

Now worker1 should show:

```
worker1    Ready    v1.31.14
```

```
[ubuntu@master:~$ kubectl get nodes  
NAME        STATUS    ROLES    AGE   VERSION  
master      Ready    control-plane   9d    v1.31.14  
worker1     Ready    <none>         9d    v1.31.14  
worker2     Ready    <none>         9d    v1.30.14  
ubuntu@master:~$
```

NEXT — Upgrade worker2

We repeat same exact process:

On MASTER:

```
kubectldrain worker2 --ignore-daemonsets --delete-emptydir-data --force
```

```
ubuntu@master:~$ kubectl drain worker2 --ignore-daemonsets --delete-emptydir-data --force
node/worker2 cordoned
Warning: deleting Pods that declare no controller: default/liveness-fail-pod; ignoring DaemonSet-managed
fs-csi-node-d6g6j, kube-system/kube-proxy-mz5lk
evicting pod kube-system/efs-csi-controller-6958db45ff-sm2cb
evicting pod default/liveness-fail-pod
evicting pod kube-system/calico-kube-controllers-5b9b456c66-pptv8
evicting pod kube-system/coredns-55cb58b774-4krhp
evicting pod kube-system/ebs-csi-controller-55cbf6bcd4-jnrpd
evicting pod kube-system/ebs-csi-controller-55cbf6bcd4-nfjtx
evicting pod kube-system/efs-csi-controller-6958db45ff-8v65q
error when evicting pods/"ebs-csi-controller-55cbf6bcd4-jnrpd" -n "kube-system" (will retry after 5s)
pod/efs-csi-controller-6958db45ff-8v65q evicted
pod/liveness-fail-pod evicted
pod/ebs-csi-controller-55cbf6bcd4-nfjtx evicted
pod/calico-kube-controllers-5b9b456c66-pptv8 evicted
pod/efs-csi-controller-6958db45ff-sm2cb evicted
evicting pod kube-system/ebs-csi-controller-55cbf6bcd4-jnrpd
error when evicting pods/"ebs-csi-controller-55cbf6bcd4-jnrpd" -n "kube-system" (will retry after 5s)
pod/coredns-55cb58b774-4krhp evicted
evicting pod kube-system/ebs-csi-controller-55cbf6bcd4-jnrpd
error when evicting pods/"ebs-csi-controller-55cbf6bcd4-jnrpd" -n "kube-system" (will retry after 5s)
evicting pod kube-system/ebs-csi-controller-55cbf6bcd4-jnrpd
error when evicting pods/"ebs-csi-controller-55cbf6bcd4-jnrpd" -n "kube-system" (will retry after 5s)
evicting pod kube-system/ebs-csi-controller-55cbf6bcd4-jnrpd
pod/ebs-csi-controller-55cbf6bcd4-jnrpd evicted
```

STEP 2 — Change Repo to 1.31

```
echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /"
| sudo tee /etc/apt/sources.list.d/kubernetes.list
```

Update:

```
sudo apt-get update
```

```
ubuntu@worker2:~$ echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /" | sudo tee /etc/apt/sources.list.d/kubernetes.list
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /
ubuntu@worker2:~$ sudo apt-get update
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1800 kB]
Get:7 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [331 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.5 kB]
```

STEP 3 — Upgrade kubeadm

```
sudo apt-mark unhold kubeadm
sudo apt-get install -y kubeadm=1.31.14-1.1
sudo apt-mark hold kubeadm
```

```
ubuntu@worker2:~$ sudo apt-mark unhold kubeadm
sudo apt-get install -y kubeadm=1.31.14-1.1
sudo apt-mark hold kubeadm
Canceled hold on kubeadm.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be upgraded:
  kubeadm
1 upgraded, 0 newly installed, 0 to remove and 64 not upgraded.
Need to get 11.6 MB of archives.
After this operation, 8618 kB of additional disk space will be used.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core: kubeadm 1.31.14-1.1 amd64 [11.6 MB]
Fetched 11.6 MB in 0s (52.8 MB/s)
(Reading database ... 71875 files and directories currently installed.)
Preparing to unpack .../kubeadm_1.31.14-1.1_amd64.deb ...
Unpacking kubeadm (1.31.14-1.1) over (1.30.14-1.1) ...
Setting up kubeadm (1.31.14-1.1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kubeadm set on hold.
```

Check version:

```
kubeadm version
```

It must show:

```
v1.31.14
```

```
ubuntu@worker2:~$ kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"31", GitVersion:"v1.31.14", GitCommit:"f38478c547e9da1e53b78886662370b000000000", Compiler:"gc", Platform:"linux/amd64"}
ubuntu@worker2:~$
```

i-0c8a37205a734d495 (k8s-worker2)

STEP 4 — Run kubeadm upgrade node

```
sudo kubeadm upgrade node
```

Wait until it completes.

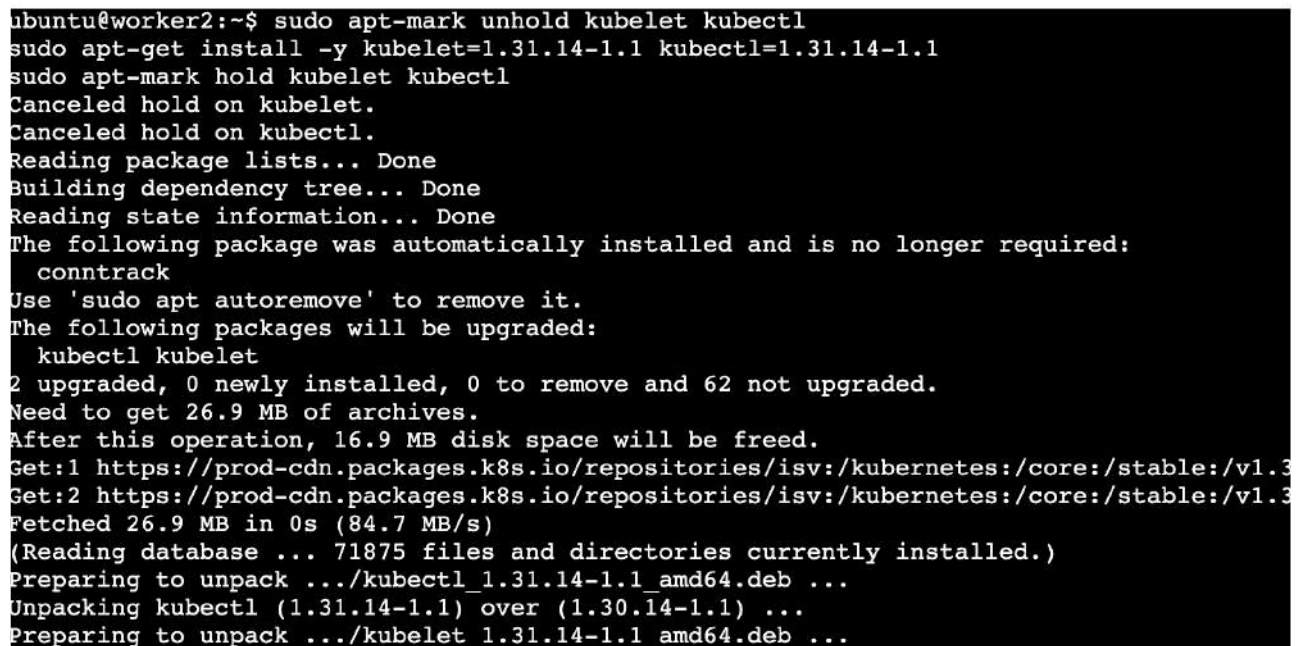
```
ubuntu@worker2:~$ sudo kubeadm upgrade node
[upgrade] Reading configuration from the cluster...
[upgrade] FYI: You can look at this config file with 'kubectl -n kube-system get configmap kubeadm-config --output yaml'
[preflight] Running pre-flight checks
[preflight] Skipping prepull. Not a control plane node.
[upgrade] Skipping phase. Not a control plane node.
[upgrade] Skipping phase. Not a control plane node.
[upgrade] Backing up kubelet config file to /etc/kubernetes/tmp/kubeadm-kubelet-backup-20240322000000
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[upgrade] The configuration for this node was successfully updated!
[upgrade] Now you should go ahead and upgrade the kubelet package using your package manager.
ubuntu@worker2:~$
```

i-0c8a37205a734d495 (k8s-worker2)

PublicIP: 3.23.92.220 PrivateIP: 172.31.0.220

STEP 5 — Upgrade kubelet & kubectl

```
sudo apt-mark unhold kubelet kubectl
sudo apt-get install -y kubelet=1.31.14-1.1
kubectl=1.31.14-1.1
sudo apt-mark hold kubelet kubectl
```

A terminal window screenshot showing the execution of apt commands to upgrade kubelet and kubectl. The output shows that the packages were successfully upgraded from version 1.30.14-1.1 to 1.31.14-1.1. The terminal text is as follows:

```
ubuntu@worker2:~$ sudo apt-mark unhold kubelet kubectl
sudo apt-get install -y kubelet=1.31.14-1.1 kubectl=1.31.14-1.1
sudo apt-mark hold kubelet kubectl
Canceled hold on kubelet.
Canceled hold on kubectl.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  conntrack
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  kubectl kubelet
2 upgraded, 0 newly installed, 0 to remove and 62 not upgraded.
Need to get 26.9 MB of archives.
After this operation, 16.9 MB disk space will be freed.
Get:1 https://prod-cdn.packages.k8s.io/repositories/iscv:/kubernetes:/core:/stable:/v1.31.14-1.1/kubectl_1.31.14-1.1_amd64.deb 26.9 MB
Get:2 https://prod-cdn.packages.k8s.io/repositories/iscv:/kubernetes:/core:/stable:/v1.31.14-1.1/kubelet_1.31.14-1.1_amd64.deb 26.9 MB
Fetched 26.9 MB in 0s (84.7 MB/s)
(Reading database ... 71875 files and directories currently installed.)
Preparing to unpack .../kubectl_1.31.14-1.1_amd64.deb ...
Unpacking kubectl (1.31.14-1.1) over (1.30.14-1.1) ...
Preparing to unpack .../kubelet_1.31.14-1.1_amd64.deb ...
```

Restart kubelet:

```
sudo systemctl daemon-reload
sudo systemctl restart kubelet
```

When done, type:

DONE WORKER2

```
ubuntu@worker2:~$ sudo systemctl daemon-reload
sudo systemctl restart kubelet
ubuntu@worker2:~$
```

Back to MASTER

After worker2 steps complete:

```
kubect1 uncordon worker2
kubect1 get nodes
```

Final Expected Output

```
master      v1.31.14
worker1     v1.31.14
worker2     v1.31.14
```

```
node/worker2 uncordoned
ubuntu@master:~$ kubect1 uncordon worker2
node/worker2 uncordoned
ubuntu@master:~$ kubect1 get nodes
NAME          STATUS    ROLES          AGE    VERSION
master        Ready     control-plane  9d     v1.31.14
worker1       Ready     <none>         9d     v1.31.14
worker2       Ready     <none>         9d     v1.31.14
ubuntu@master:~$
```

